Technology Stack

Date	7 July 2026
Team ID	SWTID-2026-3702
Project Name	**A Comprehensive Measure of Well-Being**
Maximum Marks	2 Marks

Technology Stack Details

Technologies chosen for each component with justification:

S.No	Layer / Component	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3 (Inline Styles)	Used for creating responsive, visually appealing pages (home.html, indexnew.html, resultnew.html). Pure HTML/CSS ensures lightweight deployment without additional frontend frameworks.
2	Backend / Server-Side	Python + Flask 3.x	Flask is a lightweight WSGI framework ideal for deploying ML models. It handles URL routing, POST request processing, and template rendering with minimal overhead.
3	Machine Learning	Scikit-learn (LinearRegression)	Scikit-learn provides a simple and efficient implementation of Linear Regression. Chosen for its high accuracy (R-squared = 93.74%) and ease of integration with Pickle for model saving.
4	Data Manipulation	Pandas + NumPy	Pandas is used for loading the CSV dataset, preprocessing, and feature selection. NumPy handles array operations and numerical computations required by the model.
5	Data Visualization	Matplotlib + Seaborn	Matplotlib provides base plotting engine. Seaborn adds high-level statistical plots (strip plots, heatmaps) for EDA analysis during model development.
6	Model Serialization	Python Pickle (built-in)	Pickle serializes the trained LinearRegression object to HDI.pkl. Enables model reuse in Flask without retraining, saving computation time.
7	Development IDE	PyCharm / Jupyter Notebook	Jupyter Notebook used for iterative ML development and EDA. PyCharm used for Flask app development with code analysis and debugging tools.
8	Environment Manager	Anaconda Navigator	Manages Python environments, package versions, and dependencies. Ensures consistent setup across team members' machines.

9	Version Control	Git + GitHub	Used for code versioning, collaboration, and project submission. Follows structured repository layout with 8 milestone folders.
10	Cloud / Hosting	Localhost (port 5000)	Application runs locally via app.run(debug=True, port=5000). Can be deployed to Heroku or AWS in future phases.